GGG-W-686E
30 May 1994
SUPERSEDING
GGG-W-686D
AMENDMENT-2
16 AUG 1989

FEDERAL SPECIFICATION

WRENCH, TORQUE, UNIDIRECTIONAL

This Federal Specification, dated 28 Jul 93, prepared by the Air Force, Department of Defense, is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND TYPES

1.1 Scope. This specification covers unidirectional torque wrenches, manually adjustable to a predetermined value to apply measured torque. This value is indicated by feel impulse.

1.2 Types. Torque wrenches covered by this specification shall be of the following types, as specified (see 6.2):

Type I. Plain head (RH torquing).
Type II. Ratcheting head (RH torquing).
Type III. Plain head (LH torquing).
Type IV. Ratcheting head (LH torquing).
Type V. Changeable head.

2. APPLICABLE DOCUMENTS

2.1 Government Specifications and Standards. The following documents, of the issue in effect on the date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications and Standards. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the General Services administration, Specifications unit (WFSIS), 7th and D Street SW, Washington, DC 20407.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-45662 - Calibration Systems Requirements

(Copies of Military Specification and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120).

2.2 Other Publications. The following documents from a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply:

American National Standard (ANSI):

ANSI B107.4-88 - Driving and Spindle Ends for Portable Hand, Air, and Electric Tools.
ANSI B107.10M-88 - Handles and Attachments for Hand Sockets Wrenches - Inch and Metric Series.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

ANSI/UL 94 - Tests for Flammability of Plastic Materials for Parts in Devices and Appliances, Standard for

(Revisions of this standard will be made by issuing revised or additional pages bearing their date of issue. A UL Standard is current only if it incorporates the most recently adopted revisions, all of which are itemized on the transmittal notice that accompanies the latest published set of revision pages. Information concerning the latest dates of issuance of these pages is available from Underwriters Laboratories by calling 708-272-8800, Ext. 2234).

American Society for Testing and Materials (ASTM) Standard:

ASTM E 18-84 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Standard Methods of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 First article sample. Unless otherwise specified (see 6.2), two wrenches of each type and drive size shall be submitted for examination and testing in accordance with the requirements of 4.3.2.1 prior to regular production. The tests shall be performed at a government testing facility, and failure of any test shall be cause for rejection of the sample.

3.2 Material. The material to be used in the construction of torque wrenches shall be of uniform quality and construction, and shall conform in composition, heat treatment, and suitability to the standard practices of manufacturers producing equipment of the type called for in this specification. Where handgrips or other parts of the wrenches are made of plastic or rubber, they shall be shatterproof (see 4.4.5). All plastic or rubber parts shall be non-toxic and flame resistant to classes 94V-0, 94V-1 or 94V-2 in accordance with UL-94.

3.2.1 Working parts. Working parts shall be suitably designed and processed to withstand ordinary service without showing excessive wear.

3.3 Measuring element. Measuring elements used in the construction of the wrenches shall be of high quality to give extended service and provide repeat accuracy. The measuring element shall be designed to indicate the correct torque when a load is applied to the center of the hand grip.

3.4 Wrench housing (barrel), heads, handgrips, and other parts. The wrench housing, cover, heads, handgrips, and other parts shall be capable of withstanding the torque stresses imposed in the test of 4.4.1 and 4.5 without permanent set or breakage. When handgrips are of hollow construction, the walls shall not take permanent set, crack, or break during normal use, or on being subjected to the tests of 4.4.2, 4.4.5, and 4.4.6. Metal handles shall be of one-piece design or a multipiece design using a threaded or other mechanically fitted joint and shall have a suitable portion of the handle knurled, grooved, ribbed provided with a rubber covering or other similar means so as to facilitate turning of the adjustment handle. Welded construction of the head to the tang is not acceptable.

3.5 Drive requirements. Drive size shall be as specified (see 6.2). The drive size as related to the maximum capacity shall be in accordance with TABLE I. The torque wrenches shall be provided with a square drive with spring loaded steel ball in accordance with the dimensional requirements of ANSI B107.4, except hardness shall be within the range of Rockwell C 34 minimum, to C 54 maximum. The opposite flats of the square drive shall be parallel within 5 degrees of the longitudinal axis of the handle of plain head wrenches. Drive tangs shall not be made of powdered metal. Ball detent shall be located as shown in FIGURE 1.

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TORQUE WRENCH CAPACITY

Drive Size		Maximum Load		
in	mm	in-lb	ft-lb	N-m
1/4	6.3	250	-	30
3/8	10	1200	100	135
1/2	12.5	3000	250	340
3/4	20	8400	700	1000
1	25	-	2000	2700
1 1/2	40	-	6000	9000

TABLE I

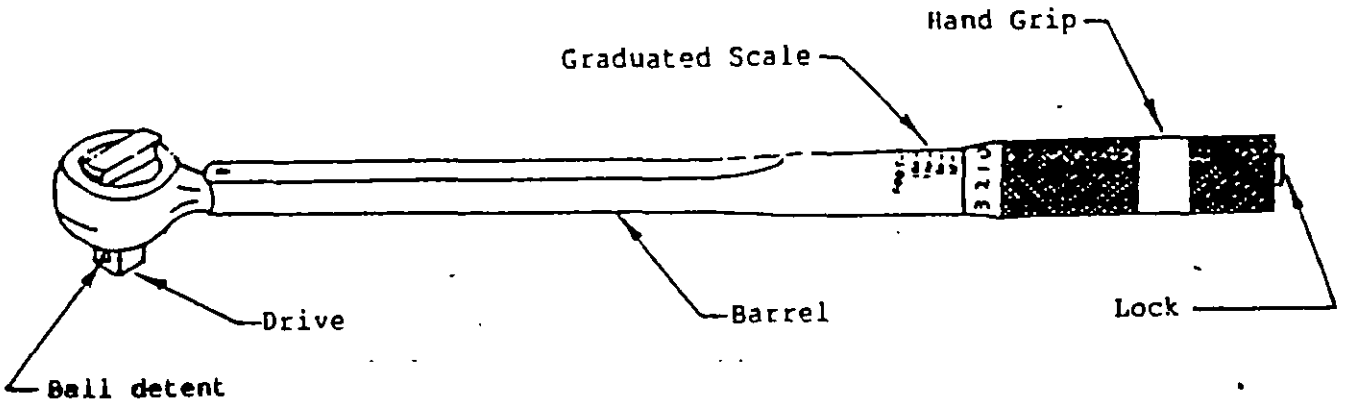


FIGURE 1

3.6 Indicator markings. Each wrench shall be marked with appropriate unit of measurement for the torque scale. The torque scale from the minimum to the maximum value shall be divided into equal graduations as indicated and numbered as appropriate. The scale graduations and identifying figures shall be cut, etched or stamped causing permanent deformation of the metal without damage to the finish coatings. No wrench shall be marked below 20 percent of the maximum capacity.

3.7 Maximum capacity. Wrenches shall be furnished in capacities as specified (see 6.2). Wrenches less than 600 foot-pound capacity shall be of such length that a maximum force of 150 pounds when applied to the center of the hand grip or to a handle extender shall be sufficient to apply the maximum rated torque without changing the specified accuracy.

3.8 Accuracy and direction

3.8.1 Accuracy. The wrench shall be accurate to within plus or minus four (4) percent of the set value, unless otherwise specified (see 6.2). The wrenches shall be capable of indicating specified torque with no more than six (6) initial excitations when tested in accordance with 4.4.

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RATED TORQUE CAPACITY (MINIMUM-MAXIMUM)		DRIVE SIZE	INCREMENT GRADUATIONS	TYPES				
In-lbs	Ft-lbs	Inch		I	II	III	IV	V
10-50	------	1/4	1 in-lb	X	X	X	X	X
30-150	------	1/4	2 in-lb	X	X	X	X	
30-150	------	3/8	1 in-lb	X	X	X	X	X
40-200	------	1/4	1 in-lb	X	X	X	X	
40-200	------	1/4	2 in-lb	X	X	X	X	
40-200	------	3/8	1 in-lb	X	X	X	X	
100-600	------	3/8	5 in-lb					X
150-750	------	3/8	5 in-lb	X	X	X	X	
200-1000	------	3/8	5 in-lb	X	X	X	X	
200-1200	------	3/8	10 in-lb					X
300-1800	------	1/2	10 in-lb					X
360-1800	------	1/2	10 in-lb	X	X	X	X	
500-2500	------	1/2	10 in-lb	X	X	X	X	
600-3600	------	1/2	10 in-lb	X	X	X	X	
700-1600	------	1/2	10 in-lb	X	X	X	X	
1200-4800	------	3/4	40 in-lb	X	X	X	X	
----------	15-75	3/8	1/2 ft-lb	X	X	X	X	
----------	16-80	3/8	1/2 ft-lb	X	X	X	X	
----------	16-80	1/2	1/2 ft-lb	X	X	X	X	
----------	20-100	3/8	1/2 ft-lb	X	X	X	X	X
----------	30-150	1/2	1 ft-lb	X	X	X	X	X
----------	40-200	1/2	1 ft-lb	X	X	X	X	
----------	50-250	1/2	1 ft-lb	X	X	X	X	
----------	50-300	3/4	2 1/2 ft-lb					X
----------	50-400	3/4	5 ft-lb					X
----------	60-300	3/4	5 ft-lb	X	X	X	X	
----------	100-500	3/4	5 ft-lb	X	X	X	X	
----------	120-600	3/4	5 ft-lb	X	X	X	X	
----------	140-700	1	2 ft-lb	X	X	X	X	
----------	500-2000	1	10 ft-lb	X		X		

X = Available in this class

TABLE II. Micrometer Torque Wrench

3.8.2 Direction. Unless otherwise specified (see 6.2), type I and type III wrenches (plain head) shall indicate torque in only one direction, as applicable (see 1.2), and shall not release or signal predetermined torque in the opposite direction. Type II and type IV wrenches (ratcheting head) shall be capable of producing torque in only one direction, as applicable (see 1.2), and shall free wheel in the opposite direction. Type V wrenches (changeable head) shall indicate torque in only one direction relative to the wrench body. The wrenches shall not release or signal predetermined torque in the opposite direction. The applied torque may be clockwise or counterclockwise, depending on the position of the head.

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3.8.3 Recalibration. The torque wrenches shall be capable of being recalibrated and adjusted in the field, using available government test equipment to the accuracy requirements of 3.8.1.

3.8.3.1 Calibration retention. The accuracy after recalibration shall be maintained (not require recalibration) for a minimum time period of 180 days under normal use as tested in 4.4.2.

3.8.4 Positional sensitivity. Wrenches shall indicate torque accurately within tolerances specified when torque is applied in accordance with 4.4.7.

3.9 Finish and coatings. All external wrench parts shall be smoothly finished. External ferrous parts, except the male tang and ball, external locking devices, screws and other fasteners, shall be protected with a chromium plated coating. All external parts shall be protected to resist corrosion. Internal parts subjected to corrosion shall be protected by the use of oil or other protective coatings. Chromium plating shall be applied in accordance with QQ-C-320, Class 1. Cadmium plating shall not be used as a protective coating.

3.9.1 Chromium plate. The coating shall be electrodeposited metals consisting of nickel (minimum thickness 0.0002 inch), or nickel and copper (minimum thickness 0.0002 inch each), followed by chrome (minimum thickness 0.000007 inch). Plating adherence and thickness shall be tested in accordance with the test requirements of QQ-C-320, Class 1.

3.10 Marking. Each wrench shall be marked in a legible and permanent manner with the manufacturer's name and/or identifying symbol, part number, and country of origin. Each wrench shall have a unique serial number consisting of a minimum of five characters. Wrenches shall also be permanently marked with the month and year of manufacture and revision letter of specification (e.g., 05-92-E, for May 1992, revision E). Marking shall be cut, etched or stamped causing permanent deformation of the metal without damage to the finish coatings.

3.10.1 Directional arrows. Each wrench shall be permanently marked with the word "TORQUE" and an arrow to indicate torque direction. Type V wrenches shall be marked as such on both sides of the barrel, so that the torquing direction is indicated when the wrench is used in either direction. Marking shall be cut, etched or stamped to the maximum size possible, causing permanent deformation of the metal without damage to the finish.

3.11 Carrying case. Each wrench shall be provided with a case. The case shall be blow molded plastic, minimum size for the particular wrench and shall fully protect the wrench from damage during shipment. The case shall be provided with a catch which cannot open without disengagement of a positive latch. It shall be close fitting to accommodate the wrench and limit movement during shipment, handling and storage. The wrench shall be accommodated in its unloaded condition (i.e., wrench set at its minimum setting). The plastic case shall be non-toxic and flame resistant to classes 94V-0, 94V-1, or 94V-2, in accordance

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with UL-94. The case shall be marked or labeled (label must be permanent and remain affixed to the case for the life of the wrench) as follows:

Wrench torque capacity _____
Manufacturer's name and model or part number _____
NSN _____
Contract number _____
Specification and revision number _____

3.12 Adjusting mechanism. The wrench shall be provided with an accurate adjusting mechanism upon which the desired torque can be hand set without the use of special tools. When the torque applied equals the set value, the wrench shall momentarily release (feel impulse) signaling that the preset torque has been reached. When the load is removed from the handle, the wrench shall automatically reset for the next operation. A positive stop shall be provided to prevent uncoupling of the adjustment mechanism.

3.12.1 Increment graduation. Minimum scale graduation shall be less than or equal to one percent of maximum capacity.

3.12.2 Accuracy calibration. The wrench shall be correctly calibrated at the manufacturer's plant in accordance with the accuracy requirements of 3.8.1. This calibration shall be performed with equipment of sufficient accuracy to comply or exceed the requirements of 4.7. This calibrating equipment must have its accuracy traceable to the National Institute of Standards and Technology in accordance with MIL-STD-45662. After calibration, the wrenches shall be protected against tampering.

3.12.3 Locking device. All wrenches shall have a positive locking device which shall hold the adjusting mechanism setting securely during the operation of the wrench. Intentional action by the operator shall be required to change the torque setting. Tools or other equipment shall not be required. If a locking collar is used, it shall be metal and shall meet the requirements of 4.4.5 and 4.4.6.

3.13 Types

3.13.1 Type I, plain head, right-hand torquing. Type I wrenches shall indicate torque in the right hand (clockwise) direction of operation and shall have a plain head. Operation shall be in accordance with 3.8.2.

3.13.2 Type II, ratcheting head, right hand torquing. Type II wrenches shall conform to the requirements for type I wrenches, except that they shall contain a ratchet mechanism enclosed in the drive head. The ratcheting device shall meet the requirements of ANSI B107.10M except that it will only produce torque in the clockwise direction.

3.13.3 Type III, plain head, left-hand torquing. Type III wrenches shall indicate torque in the left hand (counterclockwise) direction of operation and shall have a plain head. Operation shall be in accordance with 3.8.2.

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3.13.4 Type IV, ratcheting head, left hand torquing. Type IV wenches shall conform to the requirements for type III wrenches, except that they shall contain a ratchet mechanism enclosed in the drive head. The ratcheting device shall meet the requirements of ANSI B107.10M except that it will only produce torque in the counterclockwise direction.

3.13.5 Type V, changeable head. Type V wrenches shall consist of a wrench body with changeable drive head and shall indicate torque in one direction relative to the wrench body. Operation shall be in accordance with 3.8.2. Wrench may be supplied with one or more drive heads (see 6.2).

3.14 Instructions. Manufacturers shall furnish one copy of operating, calibration, and maintenance instructions with each wrench. The instructions shall include procedures concerning the use of extensions, adapters, attachments and handle extensions. The instructions shall further indicate that the wrench shall not be used below 20 percent of maximum capacity. Warranty information shall be provided as applicable.

3.15 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. The wrenches shall be free of all defects that would impair their accuracy, serviceability, durability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1. Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that material and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in Table III.

4.3 Classification of inspections. Inspections shall consist of the following examinations and tests.

4.3.1 Examinations

4.3.1.1 Visual examination. Each first article and lot acceptance sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in Table IV.

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SAMPLING DATA

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	S-4	2.5	Defects per hundred units	4.3.1.1
Dimensional examination	1 ea.	S-4	2.5	Defects per hundred units	4.3.1.2
Lot acceptance testing	1 ea.	S-1	1.0	Percent defective	4.3.2.2
Preparation for delivery	1 ea.	S-4	2.5	Defects per hundred units	4.3.3

TABLE III

4.3.1.2 Dimensional examination. Each first article and lot acceptance sample unit shall be examined for any nonconformance to dimensional requirements.

4.3.2 Testing

4.3.2.1 First article testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.5. Failure of any test shall be cause for rejection.

4.3.2.2 Lot acceptance testing. Each sample unit shall be tested in accordance with 4.6.

4.3.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of section 5. Sampling shall be in accordance with Table III.

4.4 Tests

4.4.1 Overload test. Each wrench shall be subjected to an overload of 25 percent over maximum capacity prior to additional testing.

4.4.2 Calibration retention test. The wrench shall be tested 25 times in accordance with 4.4.3 during a 30 day period. The tests shall be at random intervals and in no case shall exceed three times per day. Upon conclusion of the 30 day period, the unit shall sit idle for an additional 30 days. The wrench shall then be retested 25 times during a 30 day period in accordance with 4.4.3. A minimum of 96 percent of the readings shall be within the accuracy requirements of 3.8.1. No reading shall be more than 2 percent outside the accuracy requirements of 3.8.1.

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4.4.3. Calibration accuracy test. With no more than six initial excitations, the wrench shall be tested over the entire range of three test loads. Test loads for each size wrench shall be 20%, 60%, and 100% of full capacity. The wrench shall be secured to the tester with the handle parallel to the floor and the square drive pointing down. Load shall then be applied to the center of the handgrip. The wrench shall be tested from the low to high test loads. At the instant the signal mechanism is actuated (first peak) the test value shall be recorded. The reading shall be recorded for each test value. Each reading shall be within the limits specified in 3.8.1 except as specified in 4.4.2. This procedure shall be repeated two more times without prior excitations.

CLASSIFICATION OF DEFECTS

Categories	Major Defects
101	Type, capacity, accuracy or drive square not as specified.
102	Materials defective or not as required; evidence of cracks, deep pits, voids, heavy scale, deep tool marks, porosity, or foreign inclusions.
103	Scale illegible or not as specified; graduations not uniform, not adequately spaced, not permanent, or graduations not identified (numbered) as required.
104	Carrying-case not as specified.
105	Male tang defective, not provided with a springloaded ball or ball not staked in place.
106	Type of signal mechanism utilized for producing the torque indication not as specified.
107	Torque setting mechanism cannot be readily adjusted or setting cannot be locked.
108	Ratcheting mechanism not as specified.
109	Not provided with an adapter handle (where applicable).
110	Wrench or carrying case marking or labeling not as specified, or not permanent.
111	Instructions not provided or not as specified.
112	Wrench not smoothly finished; evidence of fins, burrs, rough surfaces or sharp edges.

TABLE IV

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4.4.4 Hardness test for square drives. Each sample wrench shall be subjected to a hardness test to verify conformance to the value specified for drive squares specified in 3.5. Test shall be in accordance with ASTM E 18-84.

4.4.5 Handgrip and locking collar shatter test. The handgrip shall be placed on a solid foundation. A steel weight with a flat surface and a weight of 5 + 0.125/ - 0.0 pounds shall be dropped directly upon the handgrip three times from a 3-foot height. If the handgrip is functional and unbroken after the three direct blows, it shall have passed the shatter test. If a locking collar is used, it shall be subjected to the above test and shall use the same failure criteria.

4.4.6 Drop test. One sample wrench shall be dropped a distance of 3 feet onto a concrete slab a total of three times; the first time the wrench shall be made to drop on the square drive end, the second time on the opposite end, and third time it shall be made to drop on its side. Breakage or damage to the wrench or its component parts to the extent that the torque wrench cannot be used or recalibrated to the specified accuracy (see 3.8.1), or parts breaking off shall be cause for rejection.

4.4.7 Positional sensitivity test. Selected torque wrenches shall be tested for positional sensitivity in the positions indicated in Figure 2. Each sample torque wrench shall be tested in the direction of indicated torque at 20, 60, and 100 percent of capacity for each position. The test load shall be applied to the wrench handle until the wrench indicates the proper torque. The tester reading shall then be recorded. Each value for a particular nominal torque shall remain within the accuracy requirements of 3.8.1 with the exception of 150 in-lb and below wrenches. The 150 in-lb and below wrenches must meet the accuracy requirements of 3.8.1 for the 60 and 100 percent of capacity torque settings and be no more than 2 percent outside the 3.8.1 requirement for the 20 percent capacity torque setting.

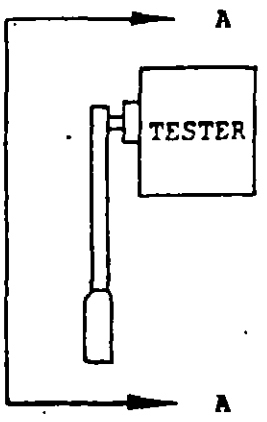
4.5 Endurance test. Each sample wrench shall be stressed from a no-load condition to a full load condition 5,000 times in the direction of indicated torque. Wrenches shall be stressed at a rate of not greater than 60 times per minute. At the conclusion of this test, without performance of maintenance or recalibration, the wrenches shall be subjected to the test of 4.4.3. If the wrench passes the test of 4.4.3, each sample wrench shall be stressed from a no-load condition to 50 percent of full load condition 20,000 times in the direction of indicated torque. At the conclusion of the 20,000 cycle test and without recalibration, the sample wrench shall be subjected to the test of 4.4.3 and shall not be more than 2 percent outside of the accuracy requirements of 3.8.1. After recalibration performed in accordance with the manufacturer's instructions (see 3.14), if required, the wrench shall be subjected to the test of 4.4.3 and shall meet accuracy requirements of 3.8.1. Deformation or physical failure at any time during the endurance test shall be cause for rejection.

4.5.1 Endurance test, Type V wrench. Type V wrenches shall be tested in accordance with 4.5, using an interchangeable head capable of withstanding the maximum capacity of the torque wrench. If provided with more than one drive head combination, each head shall be tested in accordance with 4.5 with the wrench set to the maximum capacity of the drive head. Deformation or physical failure of the drive head shall be cause for rejection for the additional head tests.

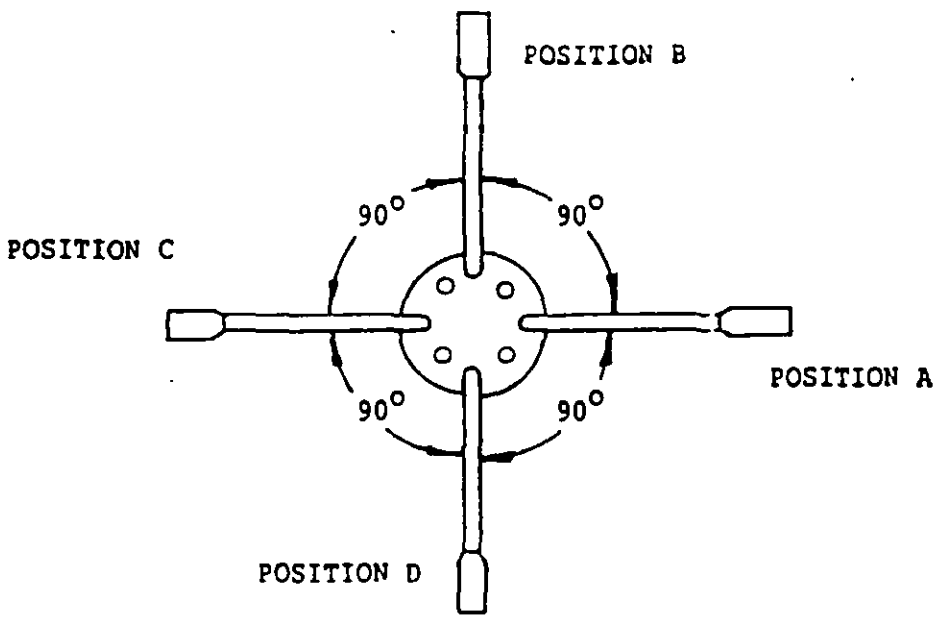
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4.6 Endurance tests for lot acceptance. For lot acceptance, the sample wrenches shall be tested as specified in 4.5 except the number of cycles for each wrench tested shall be 2,000 times in the direction of indicated torque. Without recalibration, the wrenches shall be subjected to the test of 4.4.3.

4.7 Performance tests for lot acceptance. The government reserves the right to have a government representative select samples from random lots during the contract period and send the wrenches to a government of independent laboratory for testing in accordance with 4.4.1.



VIEW A. Vertical Plane
(Handle shown in
Position D only)



VIEW B. Test Positions

FIGURE 2. Positional Sensitivity Test Positions (Section A-A).

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4.8 Test equipment. The torque wrench tester shall have an electronic digital readout, electronic transducers, and shall have an accuracy equal to or less than +/- 0.5 percent of indicated value for all values within the wrench's calibrated range. The tester shall capture the first peak of torque. Positioning of the wrench on the tester shall be in accordance with 4.4.3. The loading of the torque wrench for the last 20 percent of target value shall be at a constant rate not to exceed 5 percent of target value per second; the rate of loading for the first 80 percent may exceed the speed of the last 20 percent of target value. The tester shall be adjusted so the load is applied at the center of the handgrip. Direct hand loading of the torque wrench is not acceptable.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation and packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. These torque wrenches are designed to signal the operator when a preset torque value has been applied to a mechanical fastener. They are generally used in situations where it is difficult to read the scale or dial of a standard torque wrench, or to repetitively apply a given torque. Torque wrenches should be selected to use the upper 80 percent of their capacity for more accurate results.

6.2 Ordering data. Purchasers should select the preferred option permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of wrench required (1.2).
(c) If first article samples are not required, (see 3.1).
(d) Drive square size required (see 3.5).
(e) Torque capacity (see 3.7).
(f) If accuracy other than 4 percent is required (see 3.8.1).
(g) Torque direction (see 1.2 and 3.8.2).
(h) If type V wrench is required, number and type of drive heads required (see 3.13.5)
(i) Level of preservation, packaging, packing and marking required (see 5.1)

6.3 Warranty information. During the warranty period, authorized DoD calibration facilities are allowed to adjust torque wrenches to the plus or minus 4 percent requirement if the wrench is a maximum of 6 percent out of tolerance without voiding the warranty. If the wrench is more than 6 percent, no attempt shall be made to adjust the wrench and it shall be returned to the manufacturer for warranty repair or replacement. This action applies to operational units only and is not intended to be used as the acceptance criteria by the General Services Administration.

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MILITARY INTEREST:

Custodians

Air Force - 99
Navy - SH
Army - GL

Review Activity

Air Force - 30

User Activity

Navy - MC

PREPARING ACTIVITY

AIR FORCE - 84
(Project Number 5120-D009)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER GGG-W-686E	2. DOCUMENT DATE (YYMMDD) 94 MAY 30

3. DOCUMENT TITLE
FEDERAL SPECIFICATION - WRENCH, TORQUE, UNIDIRECTIONAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME WR-ALC/TILBA	b. TELEPHONE (Include Area Code) (1) Commercial: 912-926-3638 (2) AUTOVON: 468-3638
c. ADDRESS (Include Zip Code) 255 SECOND STREET ROBINS AFB GA 31098-1637	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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